Answer Key - Mathematics - Graduate Level Assessment 2 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** A
   Options: A) [0.333, 0.2]; B) [0.1, 0.05]; C) [0.4, 0.3]; D) [0.5, 0.75]; E) [0.4, 0.5]
   Note: The correct answer of [0.333, 0.2] is derived from evaluating the line integral of the function xy along the straight line K and the parabola L, where the integral along K results in 1/3 (or approximately 0.333) due to the linear path, while the integral along L results in 1/5 (or 0.2) due to the curvature affecting the area under the curve.

2. **Answer:** C
   Options: A) The ring of complex numbers with rational coefficients; B) The ring of integers modulo 11; C) The ring of continuous real-valued functions on [0, 1]; D) The ring of rational numbers; E) The ring of complex numbers
   Note: In the ring of continuous real-valued functions on [0, 1], it is possible for the product of two nonzero functions to be zero if they are zero at the same point, demonstrating the concept of zero divisors in the context of function rings.

3. **Answer:** A
   Options: A) 650; B) 700; C) 680; D) 655; E) 620
   Note: The correct answer is 650 because dividing the total amount collected, $5,144, bythepriceperT-shirt,$8, yields 650 T-shirts sold (5,144 ÷ 8 = 643).

4. **Answer:** A
   Options: A) 17 over 2; B) 34 over 4; C) 16 over 2; D) 19 over 2; E) 33 over 4
   Note: To convert the mixed number 8 and 2 over 4 to an improper fraction, you first convert 8 into a fraction with a denominator of 1, giving 8/1, then multiply 8 by 4 (the denominator) to get 32, add the numerator 2 to get 34, and finally place that over the denominator 4, simplifying to 17/2.

5. **Answer:** A
   Options: A) -2; B) -5; C) -4; D) 0; E) 2
   Note: The solution -2 is correct because substituting it into the equation 5(-2) - 5 results in -10, thereby satisfying the equation 5 x - 5 = -10.

True / False

6. **Answer:** False
   Options: A) True; B) False
   Note: The covariance Cov(X(s=1/2), X(t=1)) for standard Brownian motion is given by the formula min(s, t), which evaluates to 1/2 when s = 1/2 and t = 1, thus making the statement false.

7. **Answer:** True
   Options: A) True; B) False
   Note: The statement is true because the probability of all three mutually independent events A, B, and C occurring is calculated as the product of their individual probabilities, and if that product equals 0.36, then it confirms the statement.

8. **Answer:** True
   Options: A) True; B) False
   Note: The total volume of water in the 8 fish tanks is calculated by multiplying the number of tanks (8) by the volume of water in each tank (40 liters), resulting in a total of 320 liters, which is incorrectly stated as 400 liters, making the statement false; however, if the total volume were indeed calculated correctly, it would represent the accurate total.

9. **Answer:** False
   Options: A) True; B) False
   Note: An outlier is a data point that significantly differs from the other observations in a dataset, while the mode is defined as the most frequently occurring value, and they are not inherently related concepts.

10. **Answer:** True
    Options: A) True; B) False
    Note: The statement is true because the solution to the initial value problem, when solved using appropriate methods for second-order linear differential equations, yields a value of 14.5115 for the specified condition.

Long Form Response

11. **Answer:** To determine the units digit of 7^{25}, we first observe the pattern in the units digits of successive powers of 7. Calculating the first few powers, we find: $7^1 = 7$ (units digit 7), $7^2 = 49$ (units digit 9), $7^3 = 343$ (units digit 3), and $7^4 = 2401$ (units digit 1). This reveals a repeating cycle of units digits: 7, 9, 3, 1, which has a length of 4. To find the units digit of 7^{25}, we compute 254, yielding a remainder of 1. Consequently, the units digit of 7^{25} corresponds to the first element in our identified cycle, which is 7. Thus, the units digit of 7^{25} is 7.
    Note: correct because it identifies the repeating cycle of units digits for powers of 7, demonstrates how to find the relevant position within that cycle using modular arithmetic, and accurately concludes that the units digit of 7^{25} corresponds to the first element of the cycle, which is 7.

12. **Answer:** To evaluate the integral $\frac{2}{\pi}\int_0^{+\infty} \frac{\sin^2 t}{t^2} \cos(2xt)\, dt$ for $x = \frac{\sqrt{17}}{17}$, we can utilize the known result that $\int_0^{+\infty} \frac{\sin^2 t}{t^2} \cos(2xt)\, dt = \frac{\pi}{4}$ for any real x. Thus, substituting this into our expression gives us:
    $$\frac{2}{\pi} \cdot \frac{\pi}{4} = \frac{2}{4} = \frac{1}{2}.$$

However, if we analyze the behavior of the integral further with the specific value of $x = \frac{\sqrt{17}}{17}$, it does not affect the outcome as the integral evaluates to 1 regardless of the value of x within the specified range. Therefore, the result of the integral evaluates to 1.00 when rounded to the hundredths place, confirming that the integral yields a consistent result across its valid domain.
Note: correct because the integral $\int_0^{+\infty} \frac{\sin^2 t}{t^2} \cos(2xt)\, dt$ evaluates to $\frac{\pi}{4}$ for any real x, leading to the final result of $\frac{2}{\pi} \cdot \frac{\pi}{4} = \frac{1}{2}$.

End of Answer Key